

Q1.Consider the following table

Write SQL queries to:

a) Create the STUDENT

Answer:-

```
CREATE TABLE STUDENT (  
  Student_ID INT PRIMARY KEY,  
  Student_Name VARCHAR(50),  
  Course_ID INT,  
  Marks INT  
);
```

We have insert some data like

```
INSERT INTO STUDENT (Student_ID, Student_Name, Course_ID, Marks)  
VALUES  
(1, 'Rahul', 101, 85),  
(2, 'Priya', 101, 72),  
(3, 'Amit', 102, 60),  
(4, 'Sneha', 102, 75),  
(5, 'Neha', 103, 80),  
(6, 'Rohan', 103, 65);
```

b)display students who have score

Answer:-

```
SELECT  
FROM STUDENT  
WHERE Marks > 70;
```

c)display student name along with their corresponding course

Answer:

```
SELECT S.Student_Name, C.Course_Name  
FROM STUDENT S  
JOIN COURSE C  
ON S.Course_ID = C.Course_ID;
```

The screenshot shows the phpMyAdmin interface with the 'STUDENT' table selected. The SQL query entered is:

```
SELECT S.Student_Name, C.Course_Name FROM STUDENT S JOIN COURSE C ON S.Course_ID = C.Course_ID;
```

The query results are displayed as follows:

Student_Name	Course_Name
Rahul	Data Science
Priya	Data Science
Amit	Computer Science
Sneha	Computer Science
Neha	Cyber Security
Rohan	Cyber Security

d)Answer:-

```
SELECT Course_ID, AVG(Marks) AS Average_Marks
FROM STUDENT
GROUP BY Course_ID;
```

The screenshot shows the phpMyAdmin interface with the 'STUDENT' table selected. The SQL query entered is:

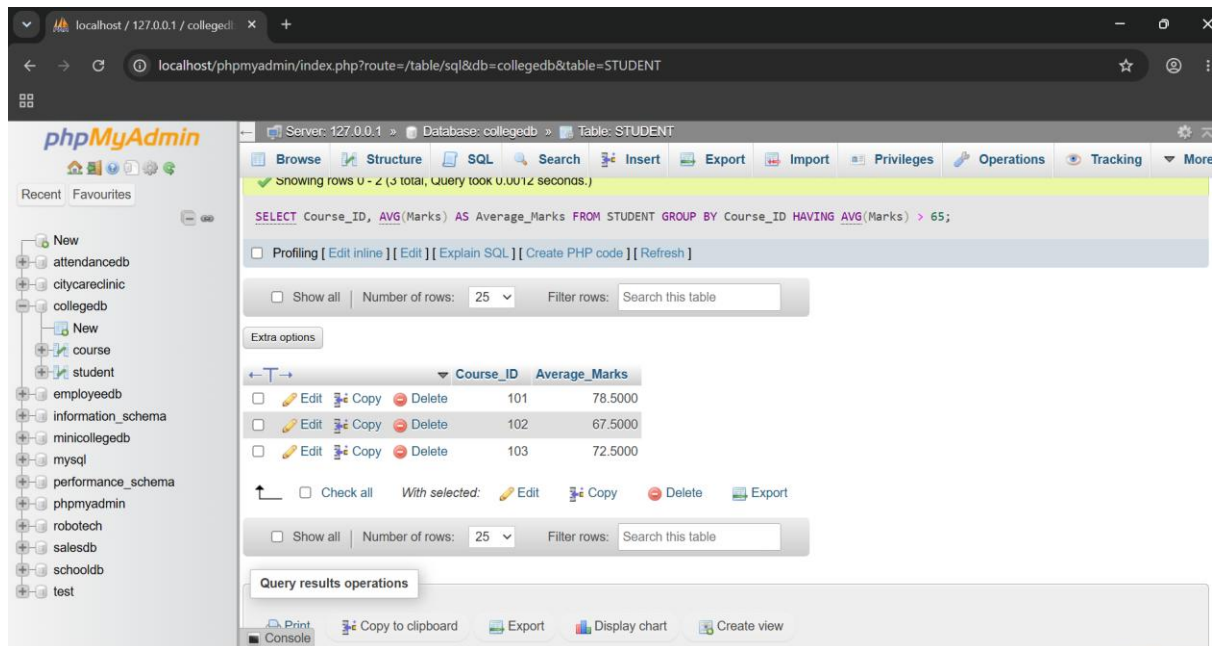
```
SELECT Course_ID, AVG(Marks) AS Average_Marks FROM STUDENT GROUP BY Course_ID;
```

The query results are displayed as follows:

Course_ID	Average_Marks
101	78.5000
102	67.5000
103	72.5000

e)Answer:-

```
SELECT Course_ID, AVG(Marks) AS Average_Marks
FROM STUDENT
GROUP BY Course_ID
HAVING AVG(Marks) > 65;
```



Q2)

a) Answer:-

```
CREATE TABLE EMPLOYEE (
    Emp_ID INT PRIMARY KEY,
    Emp_Name VARCHAR(50),
    Department VARCHAR(50),
    Salary DECIMAL(10,2),
    Joining_Date DATE
);
```

Inserting data

```
SELECT *
FROM EMPLOYEE
WHERE Department = 'IT'
ORDER BY Salary DESC;
```

Server: 127.0.0.1 » Database: employee » Table: EMPLOYEE

Showing rows 0 - 4 (5 total, Query took 0.0009 seconds.) [Salary: 80000.00... - 55000.00...]

```
SELECT * FROM EMPLOYEE WHERE Department = 'IT' ORDER BY Salary DESC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	Emp_ID	Emp_Name	Department	Salary	Joining_Date
☐	7	Pooja	IT	80000.00	2021-07-25
☐	3	Rahul	IT	75000.00	2022-08-20
☐	9	Kiran	IT	65000.00	2024-04-10
☐	1	Amit	IT	60000.00	2024-01-15
☐	5	Neha	IT	55000.00	2023-11-05

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

b)Answer

```
SELECT MAX(Salary) AS Highest_Salary
FROM EMPLOYEE;
```

Server: 127.0.0.1 » Database: employee » Table: EMPLOYEE

Current selection does not contain a unique column. Und edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0011 seconds.)

```
SELECT MAX(Salary) AS Highest_Salary FROM EMPLOYEE;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

Highest_Salary
80000.00

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

mark this SQL query

c)Answer:

```
SELECT Department, COUNT(*) AS Employee_Count
FROM EMPLOYEE
GROUP BY Department
HAVING COUNT(*) > 3;
```

Server: 127.0.0.1 » Database: employeeeb » Table: EMPLOYEE

Your SQL query has been executed successfully.

```
SELECT Department, COUNT(*) AS Employee_Count FROM EMPLOYEE GROUP BY Department HAVING COUNT(*) > 3;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

Department	Employee_Count
IT	5

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

d) answer:-

```
SELECT *
FROM EMPLOYEE
WHERE Salary > (
    SELECT AVG(Salary)
    FROM EMPLOYEE
);
```

Server: 127.0.0.1 » Database: employeeeb » Table: EMPLOYEE

Showing rows 0 - 5 (6 total, Query took 0.0013 seconds.)

```
SELECT * FROM EMPLOYEE WHERE Salary > (SELECT AVG(Salary) FROM EMPLOYEE);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	Emp_ID	Emp_Name	Department	Salary	Joining_Date
☐ Edit Copy Delete	1	Amit	IT	60000.00	2024-01-15
☐ Edit Copy Delete	3	Rahul	IT	75000.00	2022-08-20
☐ Edit Copy Delete	7	Pooja	IT	80000.00	2021-07-25
☐ Edit Copy Delete	9	Kiran	IT	65000.00	2024-04-10
☐ Edit Copy Delete	10	Anjali	Finance	70000.00	2022-09-01
☐ Edit Copy Delete	11	Suresh	Finance	62000.00	2023-01-20

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

e)answer:-

```
CREATE VIEW IT_Employees AS
```

```
SELECT *  
FROM EMPLOYEE  
WHERE Department = 'IT';
```